

Swami Sahajanand College of Computer Science
B.C.A. SEM - VI

Subject: Management Information System

Minor - 6

UNIT-1

Unit-1

Introduction to Management Information Systems and The Structure of MIS

- MIS Concepts –EIS, DSS, MRS, TPS and OIS
- Concept of Organization, Management and Information
- Information – Meaning, Uses and Cost of Information
- The need for Information system
- Types of organizational Information: TPS, MRS, DSS, EIS, OIS
- Characteristics of MRS
- Reports by MRS – Report's forms: Scheduled(Periodic) Report
- Characteristics of DSS
- Characteristics of EIS

MIS CONCEPT

Introduction:

- ❑ The Management Information System (MIS) is a concept of the last decade or two.
- ❑ It has been understood and described in a number of ways.
- ❑ It is also known as the Information System, the Information and Decision System, the Computer-based information System.

Definition:

- ❑ The MIS has more than one definition, some of which are given below.
 - ✓ The MIS is defined as a system which provides information support for decision making in the organization.
 - ✓ The MIS is defined as an integrated system of man and machine for providing the information to support the operations, the management and the decision making function in the organization.
 - ✓ The MIS is defined as a system based on the database of the organization evolved for the purpose of providing information to the people in the organization.
 - ✓ The MIS is defined as a Computer based Information System.
- ❑ Though there are a number of definitions, all of them converge on one single point, i.e., the MIS is a system to support the decision making function in the organization.
- ❑ However, in today's world MIS is a computerized business processing system generating
- ❑ Information for the people in the organization to meet the information needs of the
- ❑ Making to achieve the corporate objective of the organization.
- ❑ In any organization, small or big, a major portion of the time goes in data collection, processing, documenting it to the people. Hence, a major portion of the overheads goes into this kind of unproductive work in the organization. Every individual in an organization is continuously looking for some information which is needed to perform his/her task. Hence, the information is people-oriented and it varies with the nature of the people in the organization.

What is MIS:-

- ❑ MIS is the combination of human and computer based resources that results in the collection, storage, retrieval, communication and use of data for the purpose of efficient management of operations and for business planning.
- ❑ A system to convert data from internal and external resources into information and to communicate that information in an appropriate form to managers at all levels in all functions to enable them to make timely and effective decisions for planning, directing and controlling the activities for which they are responsible.

Advantages:

The following are some of the benefits for different types of management information systems.

- ❑ Companies are able to highlight their strengths and weaknesses due to the presence of revenue reports, employees' performance record etc. The identification of these aspects can help the company improve their business processes and operations.
- ❑ Giving an overall picture of the company and acting as a communication and planning tool.
- ❑ The availability of the customer data and feedback can help the company to align their business processes according to the needs of the customers. The effective management of customer data can help the company to perform direct marketing and promotion activities.

- ❑ Information is considered to be an important asset for any company in the modern competitive world. The consumer buying trends and behaviours can be predicted by the analysis of sales and revenue reports from each operating region of the company.

Role of MIS

- ❑ The role of the MIS in an organization can be compared to the role of heart in the body. The information is the blood and MIS is the heart. In the body the heart plays the role of supplying pure blood to all the elements of the body including the brain. The heart works faster and supplies more blood when needed. It regulates and controls the incoming impure blood, processes it and sends it to the destination in the quantity needed. It fulfills the needs of blood supply to human body in normal course and also in crisis.
- ❑ The MIS plays exactly the same role in the organization. The system ensures that an appropriate data is collected from the various sources, processed, and sent further to all the needy destinations. The system is expected to fulfil the information needs of an individual, a group of individuals, the management functionaries: the managers and the top management.
- ❑ The MIS satisfies the diverse needs through a variety of systems such as Query Systems, Analysis Systems, Modelling Systems and Decision Support Systems the MIS helps in Strategic Planning, Management Control, Operational Control and Transaction Processing.
- ❑ The MIS helps the clerical personnel in the transaction processing and answers their queries on the data pertaining to the transaction, the status of a particular record and references on a variety of documents.
- ❑ The MIS helps the junior management personnel by providing the operational data for planning, scheduling and control, and helps them further in decision making at the operations level to correct an out of control situation.
- ❑ The MIS helps the middle management in short term planning, target setting and controlling the business functions. It is supported by the use of the management tools of planning and control.
- ❑ The MIS helps the top management in goal setting, strategic planning and evolving the business plans and their implementation.
- ❑ The MIS plays the role of information generation, communication, problem identification and helps in the process of decision making. The MIS, therefore, plays a vital role in the management,
- ❑ administration and operations of an organization

1.1 Management Information Systems (MIS) – EIS, DSS, MRS, OIS

There are following systems in MIS

1. EIS/ESS 2. DSS 3. MRS 4. TPS 5. OIS

1 EIS (Executive Information System) OR ESS (Executive Support System):

- ❑ An executive information system (EIS) is intended to facilitate and support the information and decision-making needs of senior executives by providing easy access to both internal and external information relevant to meeting the day to day goals of the organization. It is commonly considered as a specialized form of decision support system (DSS).
- ❑ EIS are targeted at management needs to quickly judge the status of a business or section of business. These packages are aimed confidently at the type of business user who needs instant and up to date understanding of critical business information to aid (help) decision making.

- ❑ Generally an EIS is configured to display data such as order backlogs, open sales, purchase order backlogs, shipments, receipts and pending orders. This information can then be used to make executive decisions.

Executive Information Systems

Executive information systems support top managers with conveniently displayed summarized information, customized for them. They make a variety of internal and external information readily available in a highly summarized and convenient form. EIS are used to:

- ✓ Monitor the performance of the organization
- ✓ Assess the business environment
- ✓ Develop strategic directions for the company's future

2. DSS (Decision Support System):

1.

- ❑ A Decision Support System (DSS) is a collection of integrated software applications and hardware that form the backbone of an organization's decision making process.
- ❑ Companies across all industries rely on decision support tools, techniques, and models to help them assess and resolve everyday business questions.
- ❑ The decision support system is data-driven, as the entire process feeds off of the collection and availability of data to analyze.
- ❑ Business Intelligence (BI) reporting tools, processes, and methodologies are key components to any decision support system and provide end users with rich reporting, monitoring, and data analysis.

2.

- ❑ Decision Support Systems (DSS) are a specific class of computerized information system that supports business and organizational decision-making activities.
- ❑ A properly designed DSS is an interactive software-based system intended to help decision makers compile useful information from raw data, documents, personal knowledge, and/or business models to identify and solve problems and make decisions.
- ❑ DSS support non-routine design making for middle level management, also focus on rapidly changing, for which the procedure for arriving at solution may not be fully predefined in advance.
- ❑ DSS use internal information from TPS and MIS, often bring information from external sources, such as current stock price or product prices of companies.

3. TPS (Transaction Processing System):

- ❑ Operational managers need systems that keep track of the elementary activities and transaction of organization, such as sales, receipts, cash deposits, payroll, credit decisions and flow of material in factory.
- ❑ Transaction processing System (TPS) provide this kind of information.
- ❑ A transaction processing system is computerized system that performs and records the daily routine transaction necessary to conduct business, such as order entry, hotel reservation, payroll, employee record keeping, and shipping.
- ❑ The principal purpose of systems at this level is to answer routine questions and to track the flow of transactions through the organization.
- ❑ How many parts are in inventory? What happened to customer payment? To answer this type of questions, information generally must be easily available, current and accurate.

- ❑ Most of an organization's daily activities are recorded and processed by its **transaction processing system (TPS)**, which receives input data and converts them into output—information—intended for various users.
- ❑ Input data are called **transactions**—events that affect a business. A *financial transaction* is an economic event: it affects the firm's assets, is reflected in its accounting statements, and is measured in monetary terms.
- ❑ Sales of goods to customers, purchases of inventory from suppliers, and salaries paid to employees are all financial transactions. Everything else is a *nonfinancial transaction*. The marketing department, for example, might add some demographic data to its customer database. The information would be processed by the firm's TPS, but it wouldn't be a financial transaction.

4. MRS (Management Reporting System):

- ❑ After TPS was implemented, some organizations realized that the results produced by TPS are not suitable for higher level decision making and that the computer's capability to perform rapid calculations and logical functions could be used to produce meaningful information for management. As a result, MRS began to be developing so that managerial reports and summarized data could be produced. These reports helped managers perform their duties as well as provide middle management with statistical or summarized data for tactical level decision making.
- ❑ In general, MRS is usually used with TPS. TPS processes daily transactions, updates inventory, and keeps customer information while MRS uses the data from TPS to produce daily total sales, inventory ordering lists, and customer lists with different criteria. The output from MRS provides middle management with printed reports and inquiry capabilities to help maintain operations and management control of the enterprise. The concept of management reporting systems evolved as managers realized that computer processing could be used for more than just day-to-day transaction processing; it could also be used to produce meaningful information for management.
- ❑ Frequently, an MRS is integrated with a TPS and the input source of MRS is usually the results of TPS. **For example**, a sales transaction can be processed by using a TPS to record the sales total and customer's information.
- ❑ A MRS can further process this data to generate reports on average sales daily or fast moving items.
- ❑ Management reporting systems are the most highly structured of the management oriented information systems. The main objective of management reporting systems (MRS) is to provide lower and middle management with printed or electronic reports and with inquiry capabilities to help maintain operational and management control of the enterprise.

MRSS provide the following types of reports:

✓ Periodic Reports

- are supply on a daily, weekly, or other regular basis depending on the decision-making need.
- the format and the informational content of scheduled reports are fixed in advance. However, it is key to identify the essential informational needs of various managers to facilitate each manager's decision making and to prevent information overload.

✓ Exception Report

- An exception report is where data is not normal or not within expected parameters.
- For example, you might forecast to spend of \$400 per month on something with a variation of 20%.

- On the report generated by your computer system you could have the data within the 'normal' range displayed in black whereas an exception above or below the normal range could be displayed in red.
- Alternatively you could have the report program provide you with information that was out-with your normal expectations.
- In other words you only get told about spend when it is exceeding or likely to exceed your forecast.

✓ Demand (Ad Hoc) Reports

- The ability of a manager to request a demand report or screen output as needed enhances the flexibility of MRS use and gives the end user the capability to request the information and format that best suit his or her needs.
- Query languages provided by DBMSs make data accessible for demand reporting.

Concept of Organization, Management, Information

Organization

- ✓ The term **organization** has been defined in several ways. An *Organization* can be defined as a stable, formal social structure that uses resources and produces them to produce output. This output could be a physical product or service.
- ✓ Leavitt (1962) defined *Organization* as a specific configuration of
 - Structure
 - People
 - Task
 - Techniques
 - Structure:-
 - **Structure** describes the form of departments, hierarchy and committees. It influences the organization's efficiency and effectiveness.
- People:-
 - **People** refer to the skills, attitudes and social interaction of the members of the organization.
- Task:-
 - **Task** refers to the goals of the individual and the organization.
- Techniques:-
 - **Techniques** refer to the methodical approach used to perform tasks.
- ✓ Organizational structure thus refers to the institutional arrangements and mechanisms for mobilizing human, physical, financial and information resources at all levels of the system.
- ✓ Organization is also defined as a system incorporating a set of sub-systems (Katz and Kahn, 1978). These sub-systems are related group of activities, which are performed to meet the objectives of the organization.
- ✓ Numerous theorists have viewed organization differently. However, all definitions usually contain five common characteristics:
 - Composed of individuals and groups of individuals;
 - Oriented towards achieving common goals;
 - Differential functions;
 - Intended rational coordination, and
 - Continuity through time.
- ✓ **Types of Organizations:-**
 - At a macro level, it can be divided in three **types**
 - Extraction
 - Manufacturing including Construction, and

- Services.
- ✓ These three types have distinct features.
 - Extraction:-

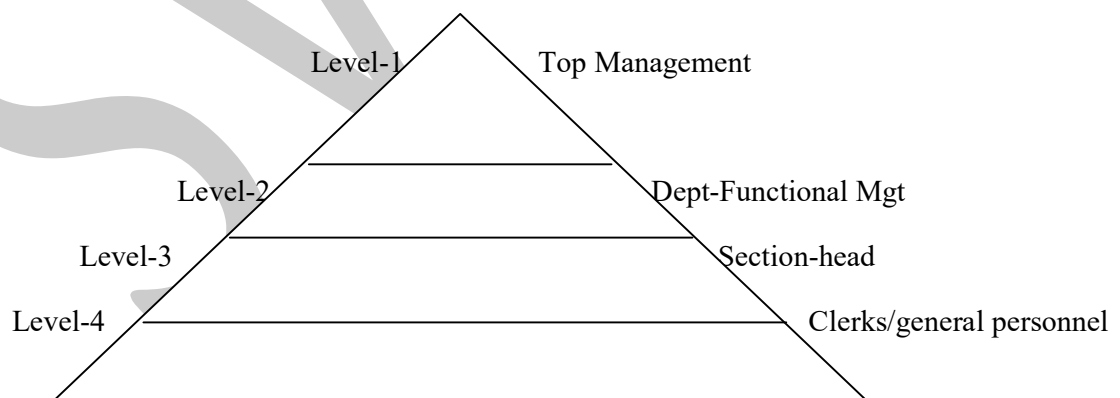
Extraction organizations produce goods by extracting them from the earth. **Examples** of such organizations are Agriculture, Mining, Oil and Gas exploration and producing companies.
 - Manufacturing including Construction:-

The manufacturing organizations produce goods by conversion or by processing of one type of goods designated as raw materials into other type of goods designated as finished goods. **Examples** of this type are Car manufacturing, Fertilizer manufacturing, pharmaceutical and drug manufacturing companies etc.
 - Services:-

Services are intangible items having no shape or color. Banking, Insurance, Transportation, Universities, Hospitals, Management Consultancy, Legal Firms etc. are **example** of these types of organizations. These three types of organizations have quite distinct structure, characteristics and functions.

Organization structure

- ✓ The formal organization is the pattern of relationships and tasks defined by official rules, policies, and system. The formal organization is designed to achieve the objectives of the organization in formal relational, efficient manner.
- ✓ An organization may contain many levels of authority or management such as, for example, in civil service, or it may have just one or two as in much small business.
- ✓ It is the number of levels between the workers and top management which determines the configuration or shape of the organization.
- ✓ Following figure shows an organization with 4 levels, typical of many small and medium sized organizations.



Structure with levels

- ✓ The organization structure is built by arrangement of organizational subsystems. The specialization, standardization organization structure is built on four basic principles, viz. hierarchy of authority, or formalization and, centralization. The structure is built to achieve goals and objectives fitting into the environment. The structure built on any principle shows

division of work, managerial and non- managerial manpower allocation as well as flows of decision responsibility and exchange of information.

- ✓ Organizations practice *division of labor* both vertically and horizontally. Vertical division includes three basic levels — top, middle, and bottom.
- ✓ The main functions of top-managers are to draw strategic plans which may include long-term and short-term goals and also to monitor the current business with the help of middle-level managers.
- ✓ Middle-level managers convert the strategic plan set by top level managers into action plan and monitor day-to-day activities of lower level managers.
- ✓ Lower level managers act as per the action plans defined by the middle level managers and perform the specific activities as assigned to them.
- ✓ **MIS : ORGANISATION**

Management information systems (MIS) should be designed, viewing the organization as discussed earlier. MIS design should give due weight-age to the human side of the organization and its culture. The task and technology are the physical aspects of the organization which can be ascertained very easily. But culture and people are very difficult to assess from the design point of view. The structure of the five sub-systems should be considered while designing the MIS.

- ✓ If the organization works on a standardized system where rules, policies, systems and procedures have been laid down, then these become part of the MIS. The processing routines in the MIS incorporate these features as an integral part. This is safe as it has already been approved by the management of the organization. Along with the information, if the decision making responsibilities are also clearly defined and allocated, then the MIS can produce information reports by processing the data and summarizing the results in line with the decision maker's position in the structure.
- ✓ The organization system is an open system and MIS should be so designed that it highlights the critical business, operational, technological and environmental changes to the concerned level in the organization, so that the action can be taken to correct the situation.
- ✓ The principle of the feed forward control should be extensively used as a design feature to provide a prior warning to the decision maker.
- ✓ Since the organization system has a dynamic role to play to meet the changing needs of a business, the MIS becomes a common support system for playing the dynamic role. When an organization is moving through the business phases of introduction, growth, maturity and decline, MIS should provide an information support, relevant to that phase.

Management

- ✓ Management is also called "Business Administration", and schools that teach management are usually called "Business Schools". The term "management" may also be used to describe the slate of managers of an organization, for example, of a corporation. A governing body is a term used to describe a group formed to manage an organization, such as a sports league.
- ✓ In for-profit work, the primary function of management is to satisfy a range of stakeholders. This typically involves making a profit (for the shareholders), creating valued products at a reasonable cost (for customers), and providing rewarding employment opportunities (for employees). In nonprofits work it is also important to keep the faith of donors. In most models of management, shareholders vote for the board of directors, and that board then hires senior management. Some organizations are experimenting with other methods (such as employee voting models) of selecting or reviewing managers/senior managers but this is very rare.
- ✓ Management is a practice not a science. It is not knowledge but performance.

Management Functions

- ✓ The five basic *Functions*, which a manager has to perform as per classical model of management, which has been accepted since 1920's and is still popular, by early authors like Henry Fayol are:

- Planning,
- Organizing,
- Coordinating,
- Directing, and
- Controlling.

To get an insight into Management let us understand these functions one by one.

▪ **Planning :**

Planning is a process of forecasting the future in advance. From where we are it leads to where we want to be. Planning requires rigorous analysis of input, output and costs. Let us suppose we want to establish a new factory or in a factory a new product.

Planning for this will require collection of basic information such as expected sales, production capacity required, types of machinery, capital, foreign exchange requirement, organization structure required to handle various jobs etc. Since planning is to chart the future course of action, it should answer the following questions and may be a few more:

- What to do?
- When to do?
- Who is to do?
- How is it to be done?
- Where is to be done?
- Why is to be done?

In an organization for facilitating this planning process, goals are set, objectives are defined, policies, rules and procedures are laid down, programs are worked out, and budgets, strategies, and schedules are finalized.

▪ **Organizing**

Organizing is the process of identifying the entire job, dividing the job into convenient subjects / tasks, allocating sub-jobs to persons or group of persons and delegating the authority to each so that the job is carried out as planned.

▪ **Coordinating**

Coordination is the process of putting the right person or group of persons at the right job. This function involves activities like defining the requirements with regard to the people for the job to be done, selecting suitable persons, training and developing them so that they can perform the desired job(s).

▪ **Directing**

Henri Fayol has identified the function of Directing as command. However, modern management experts are of the view that directing includes:

- Communication,
- Motivation, and
- Leadership.

Directing is important because in order to achieve pre-determined goals and objectives people manning the organization have to be guided, motivated and led by the manager.

▪ **Controlling :**

Controlling is the Function to ensure that things are going, as they should be. Actual performance must be compared with previously set goals. If there are any significant deviations, management must take the corrective steps to bring the things on track.

Thus controlling is the process that involves:

- Fixing standards for measuring work performance,
- Measurement of actual performance,
- Comparing actual with standards and finding out deviations, if any, and
- Taking corrective actions.

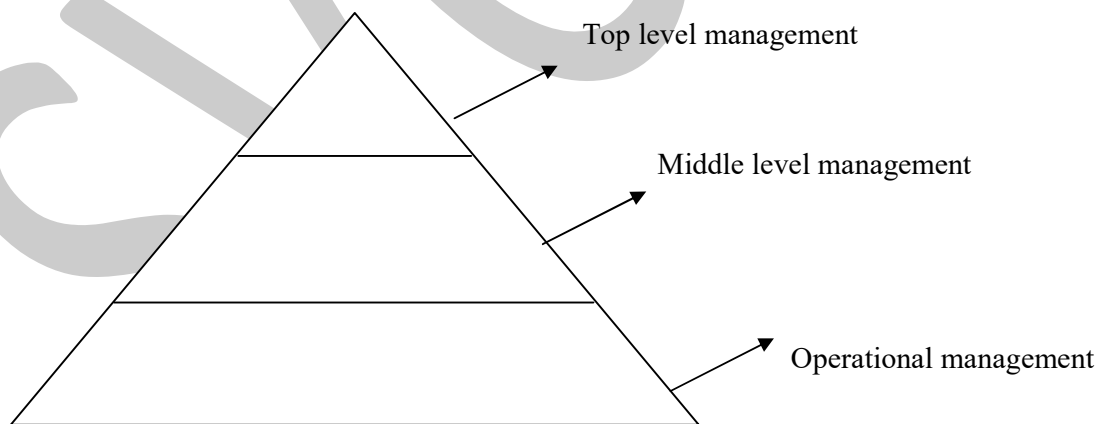
To perform these functions management has to take a variety of decisions. In other words, *decision-making* is a fundamental pre-requisite for each of the processes talked above. As per Peter Drucker 'Whatever a manager does, he does it through decision making'. Thus decision-making is the Essence of Management.

Management Functions and Business Processes

➤ Levels of Management

Managers are organizational members who are responsible for the work performance of other organizational members. Managers have formal authority to use organizational resources and to make decisions. In organizations, there are typically three levels of management: top-level, middle-level, and first-level or operational level as shown in *Figure*. These three main levels of managers form a hierarchy, in which they are ranked in order of importance.

- ✓ In most organizations, the number of managers at each level is such that the hierarchy resembles a pyramid; with many more first level managers, fewer middle managers, and the fewest managers at the top level.



Basic levels of management

Top level Managers

- ✓ Top-level managers, or top managers, are also called senior management or executives. These individuals are at the top one or two levels in an organization, and hold titles such as: Chief Executive Officer (CEO), Chief Financial Officer (CFO), Chief Operational Officer (COO), Chief Information Officer (CIO), Chairperson of

the Board, President, Vice president, Corporate head. Often, a set of these managers will constitute the top management team, which is composed of the CEO, the COO, and other department heads.

- ✓ Top-level managers make decisions affecting the entirety of the firm. Top managers do not direct the day-to-day activities of the firm; rather, they set goals for the organization and direct the company to achieve them. Top managers are ultimately responsible for the performance of the organization, and often, these managers have very visible jobs. Top managers in most organizations have a great deal of managerial experience and have moved up through the ranks of management within the company or in another firm. An exception to this is a top manager who is also an entrepreneur; such an individual may start a small company and manage it until it grows enough to support several levels of management. Many top managers possess an advanced degree, such as a Masters in Business Administration, but such a degree is not required. Some CEOs are hired in from other top management positions in other companies.
- ✓ Conversely, they may be promoted from within and groomed for top management with management development activities, coaching, and mentoring. They may be tagged for promotion through succession planning, which identifies high potential

Middle-Level Managers

- ✓ Middle-level managers, or middle managers, are those in the levels below top managers. Middle managers' job titles include: General Manager, Plant manager, Regional manager, and Divisional Manager. Middle-level managers are responsible for carrying out the goals set by top management. They do so by setting goals for their departments and other business units.
- ✓ Middle managers can motivate and assist first-line managers to achieve business objectives. Middle managers may also communicate upward, by offering suggestions and feedback to top managers. Because middle managers are more involved in the day-to-day workings of a company, they may provide valuable information to top managers to help improve the organisation's bottom line.
- ✓ Jobs in middle management vary widely in terms of responsibility and salary. Depending on the size of the company and the number of middle-level managers in the firm, middle managers may supervise only a small group of employees, or they may manage very large groups, such as an entire business location. Middle managers may be employees who were promoted from first-level manager positions within the organization, or they may have been hired from outside the firm. Some middle managers may have aspirations to hold positions in top management in the future.

First-Level / Operational Level Managers

- ✓ First-level managers are also called first-line managers or supervisors. These managers have job titles such as: Office manager, Shift Supervisor, Department Manager, Foreperson, Crew Leader, Store Manager.
- ✓ First-line managers are responsible for the daily management of line workers — the employees who actually produce the product or offer the service. There are first-line managers in every work unit in the organization. Although first-level managers typically do not set goals for the organization, they have a very strong influence on the company. These are the managers that most employees interact with on a daily basis, and if the managers perform poorly, employees may also perform poorly, may lack motivation, or may leave the company.
- ✓ In the past, most first-line managers were employees who were promoted from line positions (such as production or clerical jobs). Rarely did these employees have formal education

beyond the high school level. However, many first-line managers are now graduates of a trade school, or have a two-year associates or a four-year bachelor's degree from college.

Information:-

The word information is used commonly in our day to day working. In MIS, information has a precise meaning and it is different from data. The information has a value in decision making while data does not have. Information brings clarity and creates an intelligent human response in the mind.

In MIS a clear distinction is made between data and information. Data is like raw materials while the information is equivalent to the finished goods produced after processing the raw material. Information has certain characteristics. These are: Information

- ❑ Improves representation of an entity
- ❑ Updates the level of knowledge.
- ❑ Has a surprise value.
- ❑ Reduces uncertainty.
- ❑ Aids in decision making.

Information could be called good or bad depending on the mix of these characteristics. The sales data progressively becomes information when processed with other data such as the budget and the new product sales. Devis and Olson defines information as a data that has been processed into a form that is meaningful to the recipient and is of real or perceived value in the current or the prospective actions or decisions of the recipient. Data is defined as groups of non-random symbols in the form of text, images or voice representing quantities, actions and objects.

Information is a necessary and vital input for the Management. Any Management decision-making has to be based on information. Information need to be TRUE. Making use of its acronym we may say information must be Timely, Reliable, Useful, and Explicit.

- ❑ **Timely:** It should reach the recipient in time and up-to-date. For effective decision making information must reach the decision maker at the right time so that he can take the desired action. The delayed information at times can even lead to loss of purpose itself or in substantial loss.
- ❑ **Reliable:** Any information can be reliable only if it is factual, accurate and complete.
- ❑ **Useful:** Relevance and adequacy of the information for the need for which it is required makes it useful to purpose.
- ❑ **Explicit:** Quality information is said to be explicit if it does not require further analysis by the recipient for decision-making.

In other words information can be defined as processed data.

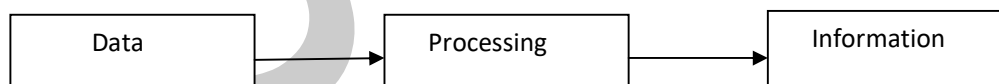


Figure 9: Information processed data System

Information System:

- ❑ In our everyday life, the word 'system' is used quite often. We talk about an educational system, computer system, organizational system, political system, solar system, railway system etc. On analysis we will find we use the word system to mean a collection of elements integrated to achieve a common definable goal. For illustration, a business organization can be considered as a system, in which the parts (departments, sections and units etc.) are joined together for a common goal. An information system can be a set of collection of elements

- starting from collection of data, its processing and presentation for achieving a defined or set goal. Data Processing Information

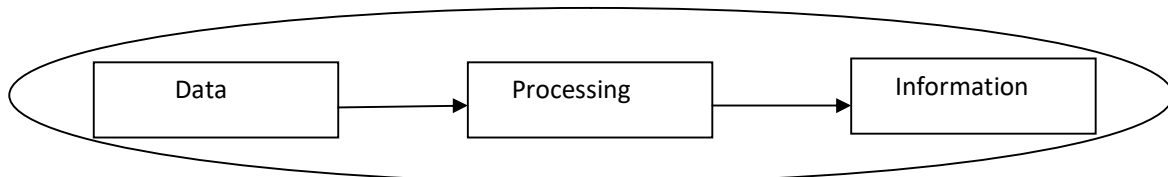


Figure 10: Information system

(In business, information systems support business processes and operations, decision-making, and competitive strategies.)

Information System is a set of processes that helps in collecting the data, storing the data, processing the data and disseminating information to support decision making process of key managers.

Use of Information

- Organizations today have access to almost unlimited amounts of data – sales, demographics, Economic trends, competitive data, consumer behaviour, efficiency measures, financial Calculations, etc.
- Businesses want information about their customers, because that information has value. We should understand what this means.
- There are two major uses of customer information that make information valuable. First, businesses use it to learn what customers want and need. Second, they use it to more efficiently offer their products to customers. Let us take each in turn.
- Businesses expend a great deal of effort to learn the desires of consumers. Almost every business is in a constant competition to please and retain customers. Each vies not only with its direct competitors, but with the makers of substitute products.
- Businesses use what information they can get to modify products, combine them with other products, and maximize their benefits and attractiveness to consumers. The more a business knows about its customers, the more able it is to satisfy them.
- The next step in the process is to tell consumers about the availability of products that may interest them. Advertising, too, is improved by information. Businesses need to learn where, when, and how their potential customers are ready to hear about products available to them. Businesses today spend an extraordinary amount of money on magazine advertisements, radio, television, billboards, and direct marketing in the hope of informing consumers about products that may meet their needs. The price of all this advertising (many billions per year across the economy) is ultimately charged to consumers.

Cost of Information

Cost of Information

The decision theory suggests the methods of solving the problems of decision making under certainty, risk and uncertainty. A decision making situation is of certainty when the decision maker has full knowledge about the alternatives and its outcomes. This is possible when perfect information is available. Therefore, the information has a perceived value in terms of decision making. The decision maker feels more secured when additional information is received in case of decision making under an uncertainty or a risk. The information is called **perfect information**, if it wipes out uncertainty or risk completely. However, perfect information is a myth. The decision theory stipulates that the value of the additional information is the value of the change in the decision behavior, resulted by the information, less the cost of obtaining the information. If the additional information does not cause any change in the decision behavior

then the value of the additional information is zero. The **value of the additional information** making the existing information perfect (VPI) is:

$$VPI = (V_2 - V_1) \cdot (C_2 - C_1)$$

Where V is the value of the information and C is the cost of obtaining the information. V_1 And C_1 relate to one set of information and V_2 , C_2 relate to the new set. If the VIP is very high, then it is beneficial to serve the additional information need.

It may be noted that the information has a value only to those who have the background knowledge to use it in a decision. The experienced manager generally uses the information most effectively but he may need less information as experience has already reduced uncertainty for him, when compared to a less experienced manager.

In MIS, the concept of the value of information is used to find out the benefit of perfect information and if the value is significantly high, the system should prove it. If the value is insignificant, it would not be worth collecting the additional information. The decisions at the operational and the middle management level are such that the value of the additional or new information is low, while at the higher levels of the management, the decision being mainly strategic and tactical in nature, the value of additional information is very high.

Apart from the monetary value of information, it has a value which is to be measured as strength in promoting the functions of the management. Some information has the strength of motivating the manager to think in futuristic terms. Some information has the strength of confirming the beliefs or understanding the business process. It also reinforces the right and wrong of a decision making process that the manager is following.

The Need for Information System

Individuals, organizations and society need to use a variety of systems to organize the collection, storage, processing, retrieval and communication of information. Personal needs tend to be simple as individuals deal with relatively small amounts of data. Organizations collect extensive amounts of information have a great need to share information among their members and so generally have more formal and extensive systems for information management than do individuals and companies use a variety of systems to satisfy these needs.

• Individual needs

- ✓ Individuals need information systems at work and at home. At work, individuals use file folders or piles on a desk to organize information stored on paper, directory systems to organize files on a computer disk or diskette and calendars to organize information about future events.
- ✓ Increasingly, individuals use laptops or other portable electronic equipment to meet their information needs. Case workers at some public housing agencies, for example, use a portable computer office, which incorporates a small computer and cellular telephone to enable case workers to enter data onsite and then transfer it later at the agency's headquarters to a larger machine for processing.
- ✓ Executives at Matsushita Electric of Canada have given portable technology to their staff and managers to allow them to work outside the traditional workplace, such as when visiting prospective and actual clients.
- ✓ At home, individuals employ systems such as posting memo notes on a refrigerator, setting alarm clock, or scheduling regular times to share information with family and friends.
- ✓ An individual who lacks such aids combined into a personal information system may miss important appointments or fail to accomplish important tasks.

- ✓ Although systems for information management by individuals generally have few components and even fewer procedures, increasingly individuals are using sophisticated personal financial managers for personal budgeting, investment tracking and bill paying.
- **Managerial Needs**
 - ✓ Most managers require significant amounts of information to perform their jobs. They typically maintain information about employee performance, customer preferences, industry trends and other subjects.
 - ✓ Increasingly managers use computerized systems to assist with obtaining, maintaining and using information, motivating, developing and communicating with other organizational members and making decisions negotiating agreements and managing resources.
 - ✓ As managers become more sophisticated in performing their tasks they require increasingly sophisticated systems to help them meet their information needs
- **Organizational Needs**
 - ✓ Because of the value of information to organizational performance most organizations develop procedures to ensure that important information is collected, captured accurately and organized effectively. Individual jobholders, various types of computer equipment and an array of computer software facilitate such information management. For example, some employees may collect information while others process it or analyze it.
 - ✓ Organizations that lack quality information systems may experience problems in accessing the data they need for executive decision making, lose important data during a relocation or power failure, perform redundant activities in dealing with customers or suppliers, or fail to respond quickly to changes in the marketplace or industry.
 - ✓ Consider a logistics agent like blue dart without a quality information system, blue dart may have massive data about package pickups but lack the procedures to translate these data into efficient systems for package delivery.
 - ✓ Many organizations even establish a special group that manages and develops information systems for the entire organization information systems also make global news and information quickly available. They allow businesses to operate internationally by making information about transactions available to managers and other workers in any country.

Information Management Model Diagnosis, Evaluation

Introduction

- ✓ An analytical model, to facilitate the effective use of information in managerial decision making.
- ✓ This model involves four steps diagnosis evaluation, design and implementation. This model has some of the same characteristics as the systems development life cycle.
- ✓ The information management model is intended to complement the systems development life cycle, the four-phase model is directed at users of information systems rather than information systems professionals or other systems designers.

Diagnosis

- ✓ Managers, employees and other individuals must begin by assessing their needs for information within a particular situation they face.

- ✓ Diagnosis requires a description of the existing problem, the context in which it occurs, the type of information available, the type of information required to solve it and the possible ways of securing the needed information.
- ✓ Diagnosis of information needs can occur at the individual, managerial or organizational levels.
- ✓ Individuals must assess their information needs at work and home.
- ✓ Managers often have needs for transaction processing, financial control, project management and communication among others.
- ✓ Organizations use information to increase their competitive advantage such as by improving customer service, cost control or quality monitoring.
- ✓ They also must identify the information they need for developing and implementing their organizational strategy.
- ✓ Society too uses information for communication, economic development and generally improving the quality of life.
- ✓ Specifying in detail the information needs at each of these levels is the first step in the effective management of information.

Evaluation

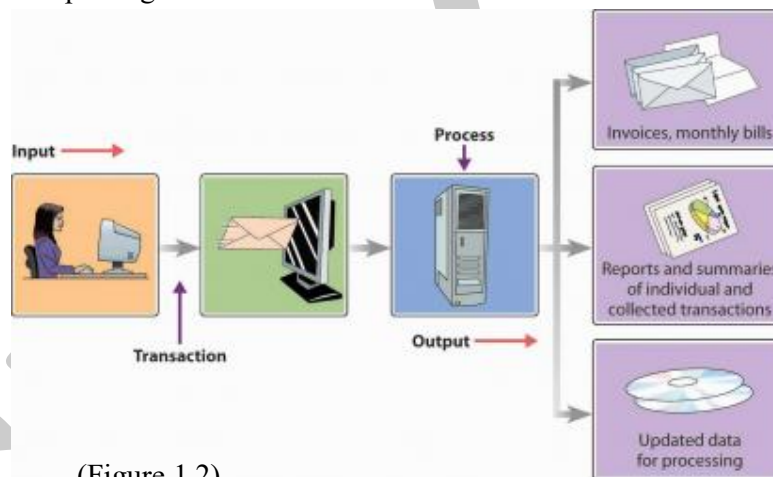
- ✓ Evaluation of the methods, techniques and systems for handling information follows the diagnosis of needs.
- ✓ This step begins with an assessment of the current manual or computerized systems for handling information.
- ✓ For example, a manager might first describe or identify the components of the information systems and technology used to acquire process, store, retrieve or communicate information.
- ✓ Next he or she might compare these components to available systems.
- ✓ How well does the current system respond to the information needs?
- ✓ Are systems available that would significantly improve the handling of information?
- ✓ What consequences will result with a change in the way information is handled?
- ✓ Finally, the manager or other user might determine what aspects of his or her information needs are not handled and which cannot be handled, regardless of the information technology or information systems used.

Type of organizational Information TPS, MRS, DSS, EIS, OIS

TPS: Transaction Processing System

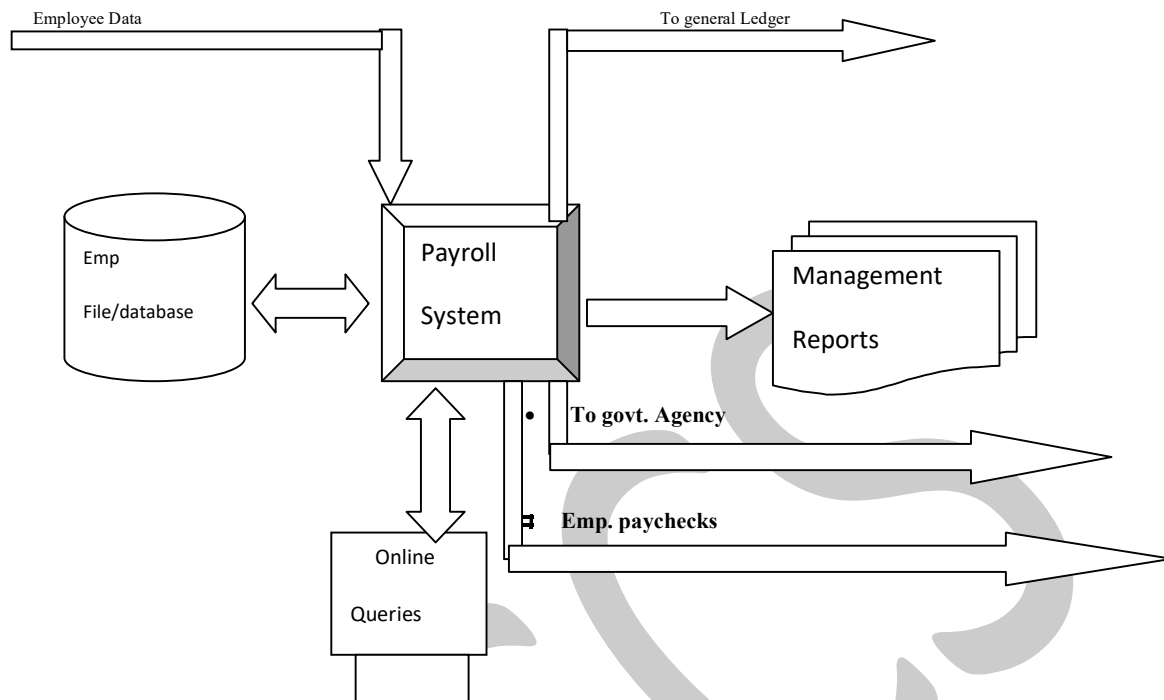
- ❑ Operational managers need systems that keep track of the elementary activities and transaction of organization, such as sales, receipts, cash deposits, payroll, credit decisions and flow of material in factory.
- ❑ A transaction processing system is computerized system that performs and records the daily routine transaction necessary to conduct business, such as order entry, hotel reservation, payroll, employee record keeping, and shipping.
- ❑ The principal purpose of systems at this level is to answer routine questions and to track the flow of transactions through the organization.

- ❑ How many parts are in inventory? What happened to customer payment? To answer this type of questions, information generally must be easily available, current and accurate.
- ❑ Most of an organization's daily activities are recorded and processed by its **transaction processing system (TPS)**, which receives input data and converts them into output—information—intended for various users.
- ❑ Input data are called **transactions**—events that affect a business. A *financial transaction* is an economic event: it affects the firm's assets, is reflected in its accounting statements, and is measured in monetary terms.
- ❑ Sales of goods to customers, purchases of inventory from suppliers, and salaries paid to employees are all financial transactions. Everything else is a *nonfinancial transaction*. The marketing department, for example, might add some demographic data to its customer database. The information would be processed by the firm's TPS, but it wouldn't be a financial transaction.
- ❑ Figure 1.2, "Transaction Processing System" illustrates a TPS in which the transaction is a customer's electronic payment of a bill. As you can see, TPS output can consist not only of documents sent to outside parties (in this case, notification of payment received), but also of information circulated internally (in the form of reports), as well as of information entered into the database for updating.



(Figure 1.2)

- ❑ Figure 1.3 shows TPS for payroll system. A payroll sys. Keep track of money paid to employees. An employee time card with the employee's name, Social security number and number of hours worked per week presents a single transaction for this system. Once this transaction is input into the system, it updates the system files, that permanently maintain employees data for organization. The data in the system are combined in different ways to create reports and send pay checks to employees.
- ❑ Following figure along with other accounting TPS, supplies data to the company's general ledger system, which is responsible for maintaining records of the firm's income and expenses and for producing reports such as income statements and balance sheets. Its also supplies employees history data for insurance, pension and other benefit calculations to firm's human resources system and employee payment data to government agencies.
- ❑ Manager needs TPS to monitor the status of internal operations and firm's relations with the external environment.



(Figure 1.3)

MRS: Management Reporting System

- ❑ After TPS was implemented, some organizations realized that the results produced by TPS are not suitable for higher level decision making and that the computer's capability to perform rapid calculations and logical functions could be used to produce meaningful information for management. As a result, MRS began to be developing so that managerial reports and summarized data could be produced. These reports helped managers perform their duties as well as provide middle management with statistical or summarized data for tactical level decision making.
- ❑ In general, MRS is usually used with TPS. TPS processes daily transactions, updates inventory, and keeps customer information while MRS uses the data from TPS to produce daily total sales, inventory ordering lists, and customer lists with different criteria. The output from MRS provides middle management with printed reports and inquiry capabilities to help maintain operations and management control of the enterprise. The concept of management reporting systems evolved as managers realized that computer processing could be used for more than just day-to-day transaction processing; it could also be used to produce meaningful information for management.
- ❑ Frequently, an MRS is integrated with a TPS and the input source of MRS is usually the results of TPS. For example, a sales transaction can be processed by using a TPS to record the sales total and customer's information. A MRS can further process this data to generate reports on average sales daily or fast moving items.
- ❑ Management reporting systems are the most highly structured of the management oriented information systems. The main objective of management reporting systems (MRS) is to provide lower and middle management with printed or electronic reports and with inquiry capabilities to help maintain operational and management control of the enterprise.

DSS: Decision Support System

Decision Support Systems are designed to help managers reach a decision by summarizing or comparing data from different resources. They are suitable for semi-structured and unstructured problems. Decision support systems often include query languages, statistical analysis capabilities, spreadsheet, and graphics to help decision makers evaluate the decision. DSS are a type of MIS expressly developed to support the decision making process. DSS facilitate a dialogue between the user, who is considering alternative problem solutions, and the systems, with its built-in models and accessible database. A typical DSS process involves retrieving a model from the model base and allocating proper data from the database.

Example of DSS

An interesting and small but powerful, DSS is estimating large American metal company that exists to carry bulk cargoes of coal, oil, and finished products for its parents company. This system calculates financial and technical details. Financial calculation includes ship/time cost (fuel, labor, capital), freight rates for various types of cargo, port expenses. Technical details include ship cargo capacity, speed, port distance, fuel and water consumption, and loading patterns (location of cargo for different ports). This system gives the answer of following questions...such as given a customer delivery schedule and what should be assigned for maximum profits?

DSS: Decision Support System

Figure 1.1 shows DSS built for this company.

Sometimes DSS is referred to as business intelligence, it helps users for better decision.

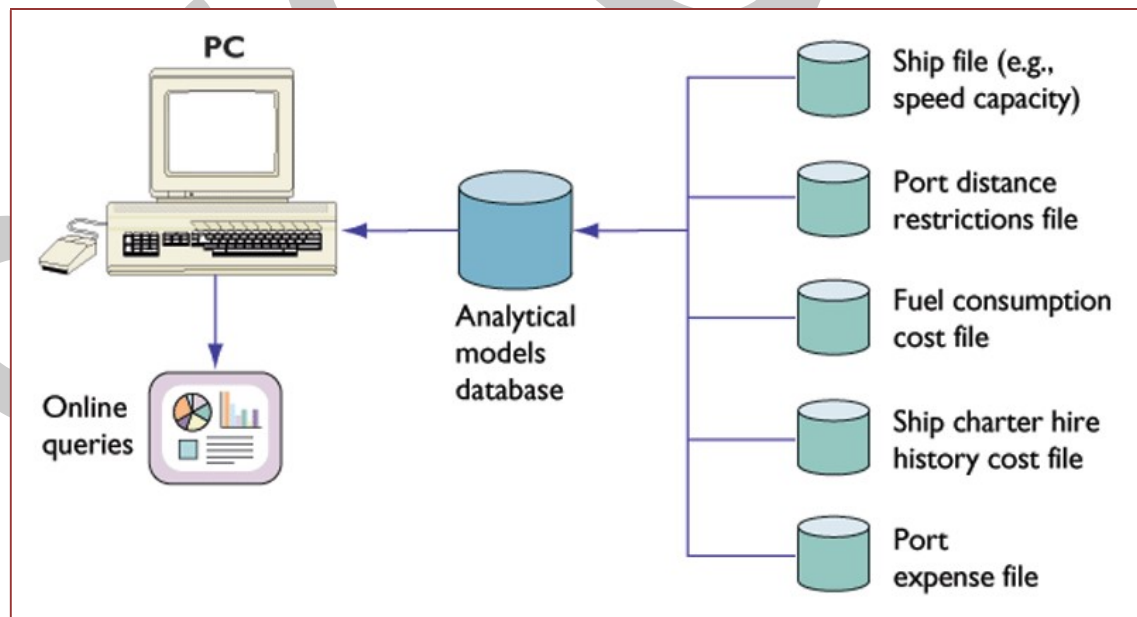


Figure 1.1

Typical information that a decision support application might gather and present would be:

- Accessing all of your current information assets, including legacy and relational data sources, cubes, data warehouses, and data marts
- Comparative sales figures between one week and the next
- Projected revenue figures based on new product sales assumptions

- ❑ The consequences of different decision alternatives, given past experience in a context that is described

✓ Support non-routine decision making for middle management.

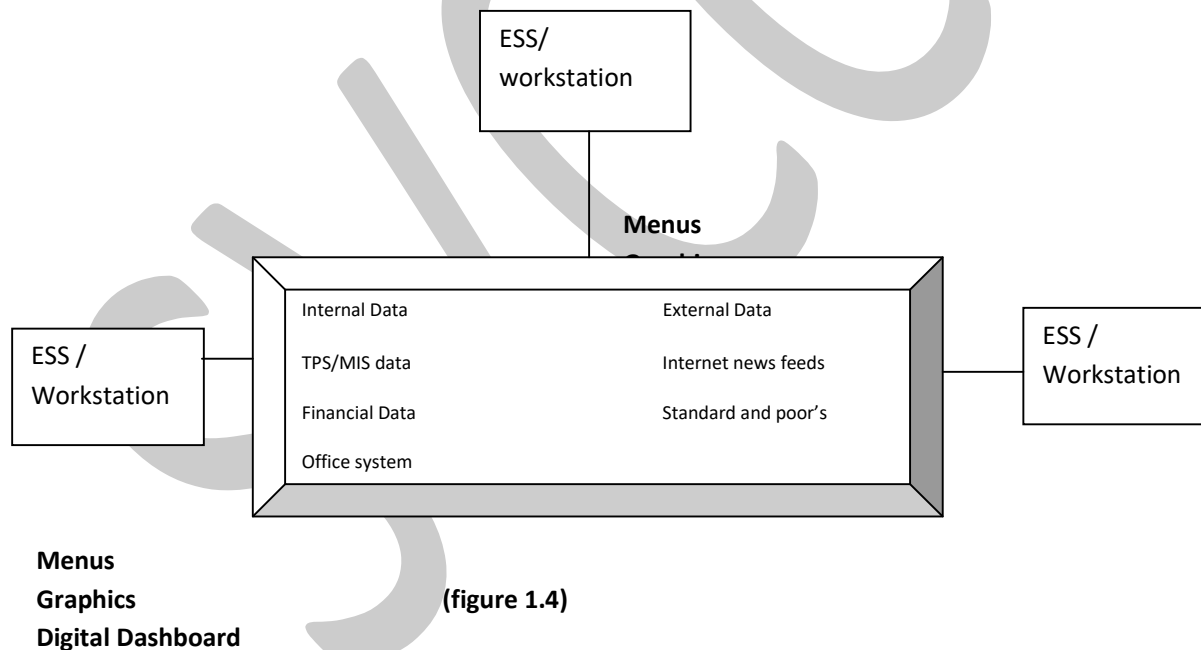
EIS: Executive Information System/ESS: Executive Support System

EIS is designed to generate information which is abstract enough to present the whole company operation in a simplified version to satisfy senior management. Characteristically, senior managers employ a great variety of informal sources of information, so that computerized information systems are able to provide only limited assistance. However, the CEO, senior and executive vice presidents, and the board of directors also need to be able to track the performance of their company and of its various units, assess the business environment, and develop strategic directions for the company's future.

Example of EIS/ESS:

ESSs are designed to incorporate data about external events, such as new tax laws or competitors, but they also draw summarized information from internal MIS and DSS. They filter, compress and track critical data, displaying the data to senior managers. For example, The CEO of Tata Steel, the largest manufacturer of steel in India, has an EIS / ESS that provides on his desktop minute-to-minute view of the firm's financial performance, and working capital, account receivable, accounts payable, cash flow and inventory. The information presents in digital Dashboard, which displays single screen graph and charts of company.

--- Figure 1.4 shows the model of ESS



OIS: Organizational Information Systems

- Identify a typical activity that would be recorded using the Transaction Processing System. How often would such an activity occur, and what member of staff would be responsible for recording it ?
- Operational management is responsible for monitoring routine daily activities. Suggest a regular weekly report that might be produced by the MIS, which includes information based on the activity you have just identified. Can you think of a situation in which this activity might be included in an exception report, in order to draw attention to a potential problem ?

- Intentional management are responsible for forward planning, based on a combination of MIS reporting and forecasting with the aid of a DSS. What sort of decision about future business operations might incorporate information about the activity you originally defined ?
- Strategic management make use of external as well as internal data in developing long term business strategies. What examples of external data might be relevant to the strategic planning of a supermarket chain ?
- Office Systems or Office Automation Systems (OAS) support general office work for handling and managing documents and facilitating communication. Text and image processing systems evolved from word processors to desktop publishing, enabling the creation of professional documents with graphics and special layout features. Spreadsheets, presentation packages like PowerPoint, personal database systems and note-taking systems (appointment book, notepad, card file) are part of OAS.
 - In addition OAS include communication systems for transmitting messages and documents (*e-mail*) and teleconferencing capabilities.

Characteristics of MRS:

1. They are usually developed by information systems professionals, rather than by end users, over an extensive period of time, with the use of life cycle oriented development methodologies as opposed to a rapid development by first building a simpler prototype system and then refining it in response to user experience.
2. These systems are build for situations in which information requirements are reasonably well known and expected to remain relatively secure.
3. MRSs do not directly support the decision-making process as a search for alternative solutions to problems and the selection of the solution to be implemented.
4. MRS are oriented toward reporting on the past and the present, rather than projecting the future.
5. MRS generally have limited analytical capabilities. They are not built around complex models, but rather rely on extraction of data from databases according to given criteria, and on summarization of the data.
6. MRS largely report on internal company operations rather than spanning the company's boundaries by reporting external information.

Reporting by Management Report Systems

MRSs may produce reports either directly from a database collected and maintained by a transaction processing system, or from databases roll off from the central database for the purpose. Separate turn off databases may be created for several reasons, such as:

1. Avoiding interference and delays in transaction processing
2. Maintaining the security of central databases
3. Economizing by using local databases accessible to local managers to counter heavy telecommunications costs of working with a central database.

MRSs provide the following types of reports:

1. Scheduled (Periodic) Reports
2. Exception Reports
3. Demand (AD hoc) Reports

1. Scheduled (Periodic) Reports:

- ✓ Periodic reports are supplied on a daily, weekly, or other regular basis depending on the decision-making need
- ✓ The format and the informational content of scheduled reports are fixed in advance. However, it is key to identify the essential informational needs of various managers to facilitate each manager's decision making and to prevent information overload.
- ✓ The concept of responsibility reporting is generally applied - managers receive reports within their specific areas of responsibility.
- ✓ A hierarchy of performance reports arises, with each report including only the items that the manager can control.

Scheduled Report Example

Daily Sales Detail Report						
						Prepared: 08/10/xx
Order #	Customer ID	Sales Rep ID	Ship Date	Quantity	Item #	Amount
P12453	C89321	CAR	08/12/96	144	P1234	\$3,214
P12453	C89321	CAR	08/12/96	288	P3214	\$5,660
P12453	C03214	GWA	08/13/96	12	P4902	\$1,224
P12455	C52313	SAK	08/12/96	24	P4012	\$2,448
P12456	C34123	JMW	08/13/96	144	P3214	\$720

2. Exception Reports:

Produced only when reestablished bounds conditions occur and containing only the information regarding these conditions. Exception reporting helps managers avoid checking of things and concentrate on unusual events.

→ Automatically produced when a situation is unusual or requires management action

Exception Report Example

Daily Sales Exception Report – ORDERS OVER \$10,000						
						Prepared: 08/10/xx
Order #	Customer ID	Sales Rep ID	Ship Date	Quantity	Item #	Amount
P12453	C89321	CAR	08/12/96	144	P1234	\$13,214
P12453	C89321	CAR	08/12/96	288	P3214	\$15,660
P12453	C03214	GWA	08/13/96	12	P4902	\$11,224
...	...	...	...	...	...	...
...	...	...	...	...	...	...

3. Demand (Ad Hoc) Reports:

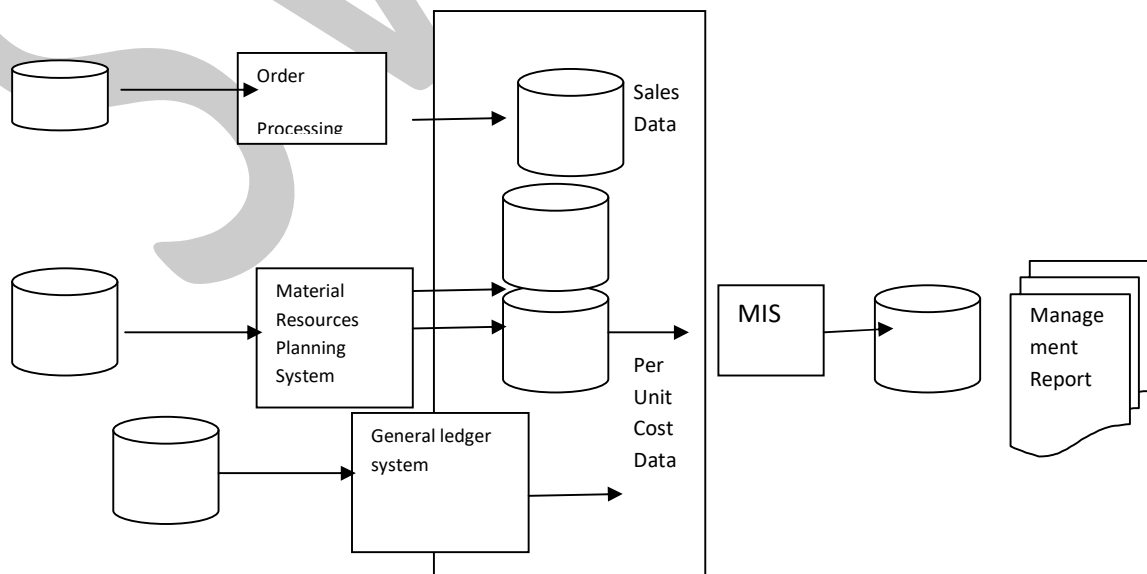
The ability of a manager to request a demand report or screen output as needed enhances the flexibility of MRS use and gives the end user the capability to request the information and format that best suit his or her needs. Query languages provided by DBMSs make data accessible for demand reporting.

Demand Report Example

Daily Sales by Sales Rep Summary Report	
Prepared: 08/10/xx	
Sales Rep ID	Amount
CAR	\$42,345
GWA	\$38,950
SAK	\$22,100
JWN	\$12,350

Management Information Systems

- Management Information Systems (MIS) generate information for monitoring performance (e.g., productivity information) and maintaining coordination (e.g., between purchasing and accounts payable). MIS extract process and summarize data from the TPS and provide periodic (weekly, monthly, quarterly) reports to managers.
- Today MIS are becoming more flexible by providing access to information whenever needed (rather than pre-specified reports on a periodic basis). Users can often generate more customized reports by selecting subsets of data (such as listing the products with 2% increase in sales over the past month), using different sorting options (by sales region, by salesperson, by highest volume of sales) and different display choices (graphical, tabular).
- Management information systems* can be defined as the study of information systems in business and management. The term *management information systems (MIS)* also designates a specific category of information systems serving management-level functions. Management information systems (MIS) serve the management level of the organization, providing managers with reports and, in some cases, with on-line access to the organization's current performance and historical records. Typically, they are oriented almost exclusively to internal, not environmental or external, events. MIS primarily serve the functions of planning, controlling, and decision making at the management level.
- Generally, they depend on underlying transaction processing systems for their data. MIS summarize and report on the company's basic operations. The basic transaction data from TPS are compressed and are usually presented in long reports that are produced on a regular schedule. *Figure 3* shows how a typical MIS transforms transaction-level data from inventory, production, and accounting into MIS files that are used to provide managers with reports. *Table 3* shows a sample report from this system.



- ✓ This diagram illustrates how management information systems obtain their data from the organization's TPS. In this system, three TPS supply summarize transaction data at the end of the time period to the MIS reporting system. Managers obtain the organisational data through the MIS, which provides them with the appropriate reports.

CHARACTERISTICS OF DSS

Attributes :

- ❑ DSS should be flexible.
- ❑ DSS should be interactive and provide ease of use.
- ❑ Effectiveness balanced with efficiency (benefit must exceed cost).
- ❑ Complete control by decision-makers.

The following are the characteristics of a DSS.

1. Facilitation. DSS facilitate and support specific decision-making activities and/or decision processes.
2. Interaction. DSS are computer-based systems designed for interactive use by decision makers or staff users who control the sequence of interaction and the operations performed.
3. Ancillary. DSS can support decision makers at any level in an organization. They are NOT intended to replace decision makers.
4. Repeated Use. DSS are intended for repeated use. A specific DSS may be used routinely or used as needed for ad hoc decision support tasks.
5. Task-oriented. DSS provide specific capabilities that support one or more tasks related to decision-making, including: intelligence and data analysis; identification and design of alternatives; choice among alternatives; and decision implementation.
6. Identifiable. DSS may be independent systems that collect or replicate data from other information systems OR subsystems of a larger, more integrated information system.
7. Decision Impact. DSS are intended to improve the accuracy, timeliness, quality and overall effectiveness of a specific decision or a set of related decisions.

DSS Supports

- ❑ Support decision makers in semi structured and unstructured problems.
- ❑ Support managers at all levels.
- ❑ Support individuals and groups.
- ❑ Support interdependent or sequential decisions.
- ❑ Support intelligence, design, choice, and implementation.
- ❑ Support variety of decision processes and styles

Benefits

1. Improves personal efficiency
2. Speed up the process of decision making
3. Increases organizational control
4. Encourages exploration and discovery on the part of the decision maker
5. Speeds up problem solving in an organization
6. Facilitates interpersonal communication
7. Promotes learning or training
8. Generates new evidence in support of a decision
9. Creates a competitive advantage over competition
10. Reveals new approaches to thinking about the problem space

11. Helps automate managerial processes
12. Create Innovative ideas to speed up the performance

Advantages, Disadvantages, Characteristic of EIS

Advantages of Executive Information System:

- 1) They provide the information that helps them make their decisions. Executives use that information, along with their experience, knowledge, education, and understanding of the corporation and the business environment as a whole, to make their decisions.
- 2) Executives want summarized data rather than detailed data (even though the details must be available). ESS rely on graphic presentation of information because it's a much quicker way for busy executives to take hold of summarized information
- 3) It provides timely delivery of company summary information.
- 4) It provides better understanding of information
- 5) It filters data for management.
- 6) It provides system for improvement in information tracking
- 7) It offers good organization to decision makers.

Disadvantages of Executive Information System:

- 1) Functions are limited, cannot perform complex calculations.
- 2) Hard to calculate benefits and to justify implementation of an EIS.
- 3) Executives may come across information overload.
- 4) System may become slow, large, and hard to manage.
- 5) Difficult to keep current data.
- 6) May lead to less reliable and unconfident data.
- 7) Small companies may not provide that much costs for implementation.

Characteristics of Successful EIS

- 1) Find an Appropriate manager Champion - EIS projects that succeed do so because at least one member of the senior management team agrees to champion the project. The executive champion need not fully understand the technical issues, but must be a person who works closely with all of the senior management team and understands their needs, work styles and their current methods of obtaining organizational information.
- 2) Deliver a Simple model Quickly - Executives judge a new EIS on the basis of how easy it is to use and how relevant the information in the system is to the current strategic issues in the organization. As a result, the best EIS projects begin as a simple prototype, delivered quickly, that provides data about at least one critical issue.
- 3) Involve Your Information Systems Department - In some organizations, the motivation for an EIS project arises in the business units quite apart from the traditional information systems (IS) organization. Consultants may be called in, or managers and analysts in the business units may take the project on without consulting or involving IS.
- 4) Communicate & Train to Overcome Resistance (*fight*)- A final characteristic of successful EIS implementations is that of communication. Executive Information Systems have the potential to hugely alter the current patterns of organizational communication and thus will typically be met with struggle. Some of this struggle is simply a matter of a lack of knowledge. Training on how to use statistics and performance measures can help.

EXPERT SYSTEM

- An expert system is an information system that captures and stores the knowledge of human experts and then imitates human reasoning and decision-making processes for those who have

less expertise. Expert systems are composed of two main components: a knowledge base and inference rules. A knowledge base is the combined subject knowledge and experiences of the human experts. The inference rules are a set of logical judgments applied to the knowledge base each time a user describes a situation to the expert system.

- ❑ Although expert systems can help decision-making at any level in an organization, non-management employees are the primary users who utilize them to help with job-related decisions. Expert systems also successfully have resolved such diverse problems as diagnosing illnesses, searching for oil and making soup.
- ❑ Expert systems are one part of an exciting branch of computer science called artificial intelligence. Artificial intelligence (AI) is the application of human intelligence to computers. AI technology can sense your actions and, based on logical assumptions and prior experience, will take the appropriate action to complete the task. AI has a variety of capabilities, including speech recognition, logical reasoning, and creative responses.
- ❑ Experts predict that AI eventually will be incorporated into most computer systems and many individual software applications. Many word processing programs already include speech recognition
- ❑ **Expert systems (ES)** are programs that mimic the judgment of experts by following sets of rules that experts would follow. They're useful in such diverse areas as medical diagnosis, portfolio management, and credit assessment.
- ❑ For example, you've called the customer-service department of your credit-card company because you want to increase your credit line. Don't expect to talk to some financial expert who's authorized to say yes or no. You'll be talking to a service representative with no financial expertise whatsoever. He or she will, however, have access to an ES, which will give you an answer in a few seconds. How does it work? The ES will prompt the representative to ask you certain questions about your salary and living expenses. It will also check internal corporate data to analyze your purchases and payment behavior, and, based on the results, it will determine whether you get an increase and, if so, how much.
- ❑ At Harrah's, an ES called the Revenue Management System helps to optimize the overall profitability of both hotel and casino operations. When a customer requests a room, the program accesses his or her profile in the database and consults certain "rules" for assessing the application. One rule, for example, might be: "If the customer has wagered more than \$100,000 in the past year, add 10 points." Eventually, the system decides whether your application will be accepted (and at what rate) by adding up points determined by the rules. While a tightwad may not get a room even when there are vacancies, a high roller may get a good rate on a luxury suite even if the hotel is nearly full.
- ❑ Expert system is understanding of fundamental principal of human experts. Its works whole work in few minutes and better than human system.

HOW EXPERT SYSTEM WORKS

- ❑ Human knowledge must work in a way in that computer can process, Expert system modeled human knowledge as a set of rules that called knowledge base.
- ❑ It has 200 to many thousands of rules, depending on complexity of problems.
- ❑ These rules are much more interconnected and nested than in a traditional software programs.
- ❑ The strategy used to search knowledge base is called inference engine. Two strategy commonly used : Forward chaining and backward chaining.

- ❑ In Forward chaining the interface engine begins with information entered by the user and searches the rules base to arrive at a conclusion.
- ❑ The strategy is to fire or the action of the rule when condition is true..
- ❑ In backward chaining The strategy searching the rules base starts with a processed by asking the user questions about selected facts either conformed or disapproved.
- ❑ Begins with right in diagram and work toward the left.

Role of Artificial Intelligence & Expert System

Today's global economy requires increased attention to the issue of business competitiveness. Business information system or Artificial Intelligence and expert system raise the competitiveness of enterprises in the global market. Business intelligence as the basis for the development and application in business information is becoming an important information technology framework that can help organization to manage, develop and communicate their intangible assets, such as information and knowledge based economy. Competency Based Management (CBM) has become vital to any firm's strategic position and organizational decision making. The role of Artificial intelligence & Expert System (ES) is to provide a knowledge based information system that is expected to have human attributes in order to replicate human capacity in ethical decision making. In this paper a holistic framework is proposed to review the ES approach that will be practically feasible for organizational settings. It will also provide executives and scholars with pragmatic understanding about integrating knowledge management strategy and technologies in business processes for successful performance. It is based on the psychological conceptions of human competence and performance in the workplace. Using an ES approach to Educational Institute with CBM will be able to more effectively use their limited resources to reap the more benefits from their investments in both people and technology. The idea of robotic domestic workers is still farfetched but companies are making progress even here. There is already a Robot Vacuum Cleaner marketed by Electrolux and doubtless improved systems with better functionality.